

SumoBot! Blue Mantis

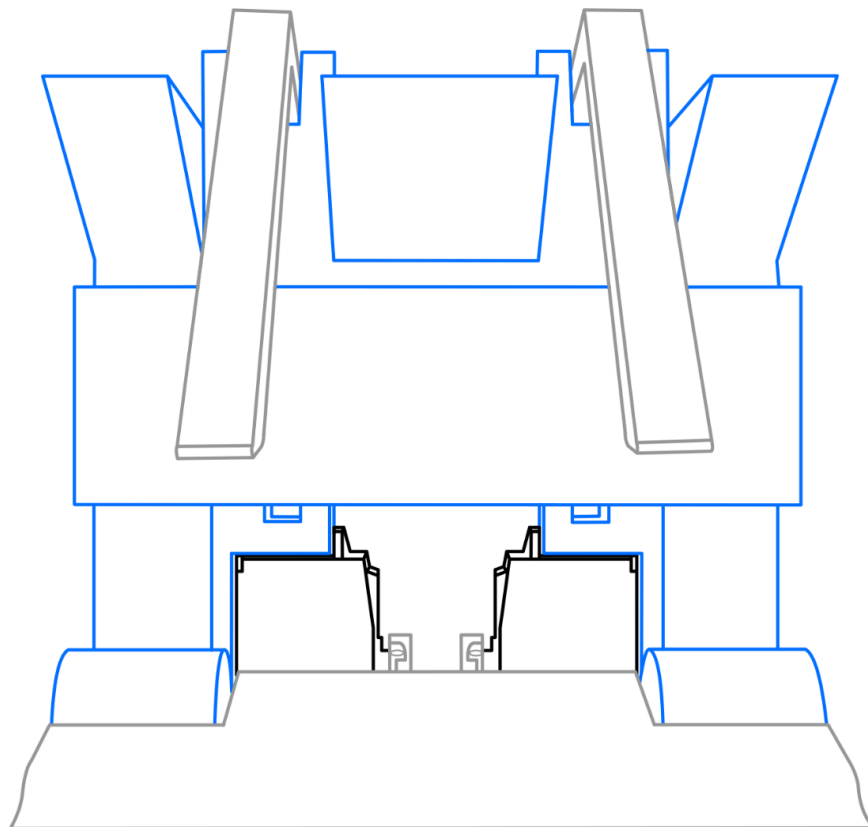
Final Report

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2026



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DISCLAIMER

This report was prepared by students as part of a university course requirement. While considerable effort has been put into the project, it is not the work of licensed engineers and has not undergone the extensive verification that is common in the profession. The information, data, conclusions, and content of this report should not be relied on or utilized without thorough, independent testing and verification. University faculty members may have been associated with this project as advisors, sponsors, or course instructors, but as such they are not responsible for the accuracy of results or conclusions.



ABSTRACT

This capstone project involves designing and building a competitive 3 kg remote-controlled sumo robot for competition at the Engineering Expo in June 2026. The robot must contain complex geometry aligning within Robogames Unified Sumobot rules[1] for a "pushing match" against competitors competing within a 154 cm diameter circular ring called a dohyo. Competition scoring follows Unified Sumo Robot rules where the first team to win two rounds receives two Yuhkoh points and wins the match.

Analysis of existing sumo robot designs[2][3] reveals common successful strategies including frontal ramps for pushing opponents, treads for increased traction, sharp angle ramps for wedging underneath opponents, and low centers of gravity for stability. Control methods typically involve either wireless RC transmitters or autonomous sensors[4]. Our sponsor relayed this project will be using a strictly remote control system with no sensors.

The goal for our project is to stay in the dohyo ring while forcing the opponent out. To achieve this, we engineered a robot with a custom offensive ramp-claw mechanism that will lift and destabilize competitors while maintaining strong forward traction. Our design emphasizes durability, controlled maneuvering, and strategic geometry to gain an advantage in close-quarters combat. Through iterative design, analysis, and testing, the project demonstrates the integration of mechanical design strategies, manufacturing feasibility, and control strategy required to produce a competitive sumobot.



ACKNOWLEDGEMENTS

The completion of this project would not have been possible without the support and guidance of many individuals and organizations. Our team would like to express sincere appreciation to the Oregon State University School of Mechanical, Industrial, and Manufacturing Engineering (MIME) for providing the resources, knowledge, and environment necessary to pursue this project. We extend special thanks to Dr. Sarah Oman, our project sponsor and primary point of contact, for her continued feedback, direction, and commitment to our team's success.

We also gratefully acknowledge Curtiss-Wright and their team of engineers for serving as our manufacturing sponsor. Their willingness to collaborate, provide machining expertise, design feedback, and fabricate our final components played a critical role in bringing our design to completion.

Finally, we thank everyone else who has offered assistance, shared knowledge, or provided encouragement throughout the duration of this project. Their contributions and support have been greatly appreciated throughout the project and do not go unnoticed.

TABLE OF CONTENTS

Contents

DISCLAIMER	1
ABSTRACT	2
ACKNOWLEDGEMENTS	3
TABLE OF CONTENTS	4
1 BACKGROUND	2
1.1 Introduction	2
1.2 Project Scope	2
2 DESIGN PROCESS	3
3 DESIGN PROPOSAL	5
4 Design Solution	8
4.1 Description of Solution	8
4.2 Project Results	9
5 LOOKING FORWARD	10
6 CONCLUSIONS-Jonah	12
7 REFERENCES	13
8 APPENDICES	14
8.1 Appendix A: Charts Utilized	14
8.2 Appendix B: Budget Sheet	18



1 BACKGROUND

1.1 Introduction

This project centers on the design and fabrication of a remote controlled sumobot, a small compact robot inspired by traditional sumo wrestling that competes in a series of organized pushing matches of three minutes with other sumobots. These robots can be autonomous[4] or remote controlled, also known as RC. Similar to human sumo wrestling, these sumobots attempt to force each other out of a circular arena, known as a dohyo. The project at hand is to create an original, strictly RC sumobot with a non-standard geometry, meaning not a box, that fits the customer requirements and Robogames Unified Sumobot rules and objective is to “fight” by applying a pushing force upon another opponent forcing them out of the dohyo ring.

The project objectives include developing practical engineering skills through iterative design, manufacturing, and testing within a team environment. Though we have found a manufacturing sponsor, our objectives still stand as we integrate mechanical and electrical systems for reliable performance and apply design principles that are feasible to be machined as we work together with a professional engineering team gaining insights within our project.

The project is sponsored by Dr.Oman and the School of Mechanical, Industrial, and Manufacturing Engineering (MIME) at Oregon State University. The project is also supported by Curtiss-Wright, the team’s manufacturing sponsor, who will produce the final mechanical components that our team has designed. Curtiss-Wright's involvement connects our academic design work with professional manufacturing practices exposing our team to industry standards and ensuring high quality fabrication.

This project is of strong interest to both sponsors because it bridges academic design and professional manufacturing. For Dr.Oman and the School of MIME, it reinforces experiential learning and innovation in mechanical design, giving our team of students hands-on exposure to robotics and fabrication. For Curtiss-Wright, it provides an opportunity to collaborate with emerging engineers and demonstrate advanced machining capabilities through the production of precision components. The project benefits all stakeholders by strengthening the connection between education and industry as we gain technical proficiency and teamwork experience, while the sponsors and university showcase the engineering and creativity that this project displays at Expo. Beyond these outcomes, the project addresses contemporary issues in robotics and automation by exploring efficient mechanical design and control strategies that mirror modern industrial automation challenges. It also promotes sustainable and efficient manufacturing practices through material and machining selection and design optimization.

1.2 Project Scope

The scope of this project is to design and fabricate an original RC sumobot that will compete in a series of “head to head” pushing matches against our competitor sumobots while staying within the Unified Sumobot rules. The problem the team is addressing is the need for a non-standard and high performing sumobot design that will demonstrate originality and mechanical efficiency while meeting customer requirements and strict competition rules. Most existing sumobots use simple box geometries; our team aims to innovate by developing a unique form in between the



rules grey areas that improves pushing power, traction, and control stability while still being within the guidelines.

Out of scope items for this project include any autonomous operation, liquids, weapons, intentional damaging mechanism, and materials that exceed weight or machinability limits. The project as stated numerous times is strictly RC, aligning with customer expectations and competition rules.

This scope represents the team's unified goal to design a competitive, manufacturable, and innovative sumobot that demonstrates engineering skills, collaboration, and real world application of design and manufacturing principles while also providing some wins during the matches at Expo.

2 DESIGN PROCESS

Our team followed a traditional product development process consisting of research, design, prototyping, and manufacturing cycles. This approach allowed us to brainstorm many design alternatives while retaining the ability to iterate based on testing results. The research phase involved analyzing previous sumo robot designs to identify successful strategies including frontal ramps, tread-based traction, sharp-angle wedging geometry, low centers of gravity for stability, and RC control reliability versus autonomous sensors.

To satisfy the requirement for at least one defensive or offensive mechanism beyond a basic ramp, the team brainstormed several supplemental features[3Autonomus]. Concepts included an aluminum bar attached to a servo for swiping opponents to the side, various lifting mechanisms inspired from battle bots, and alternative ramp geometries for improved wedging capability. The team evaluated these alternatives based on their effectiveness at neutralizing opponent resistance, manufacturability within the project timeline, and compliance with competition rules regarding push based competition without flipping.

The design proposal was based around being as innovative as we could without breaking any rules. After discussion with Dr. Oman, it was noted that there was a large grey area in the unified sumo rules that could be exploited to give an advantage. The rules are very specific about flipping, but there is no stated documentation regarding lifting the opposing bot. After discovering this, the team did a lot of research on how we could design a mechanism that could efficiently lift the opposing bot and carry it out of the ring.

The final design concept centers on a dual mechanism lifting system consisting of a bottom ramp assembly and a top clamping arm. The bottom mechanism utilizes two high torque servos mounted in parallel at the front of the chassis, controlling an angled ramp blade designed to slide underneath the opponent bot during initial contact. The innovation in this design is the integration of extruded metal support bars extending from each servo arm allowing to provide force at the front of the ramp lip. We also located a set of front wheels that have two purposes,



they allow for the bot to turn and drive more effectively, and also act as a block to provide a reaction force driven into the ground to support the ramp lifting an opposing bot.

The top mechanism consists of a servo actuated claw arm positioned above the ramp assembly. This arm engages after the ramp has successfully positioned itself under the opponent, securing the lifted bot and preventing escape. This approach provides several advantages over traditional pushing strategies. By lifting the opponent off the ground, their drive wheels lose traction, reducing their ability to resist. The clamping mechanism ensures the opponent cannot disengage, a common failure mode in ramp reliant push designs.

The drive system features four DC gear motors driving large rear wheels with wide silicone tires for maximum traction, and uses a pivot style control system to change directions quickly to maneuver around other bots. Control is implemented through an RC transmitter interfaced with an Arduino Uno, providing precise control during multiple received inputs from all subsystems. Manufacturing considerations focused on feasibility from complexity and considering weight restraints. The best combination we could optimize relies on a 3D printed chassis using standard PLA, a ramp that was created using pressed and machined aluminum, a claw that was pressed from aluminum, and lastly a servo moment arm 3D printed in a SLA resin bath.

3 DESIGN PROPOSAL

This chapter details the prototyping design process used to develop the robot, including key design considerations, iterative modeling, and physical testing. It highlights how performance priorities such as traction, stability, and simplicity informed design decisions, while also introducing the evolution of both passive and active mechanisms through successive prototypes.

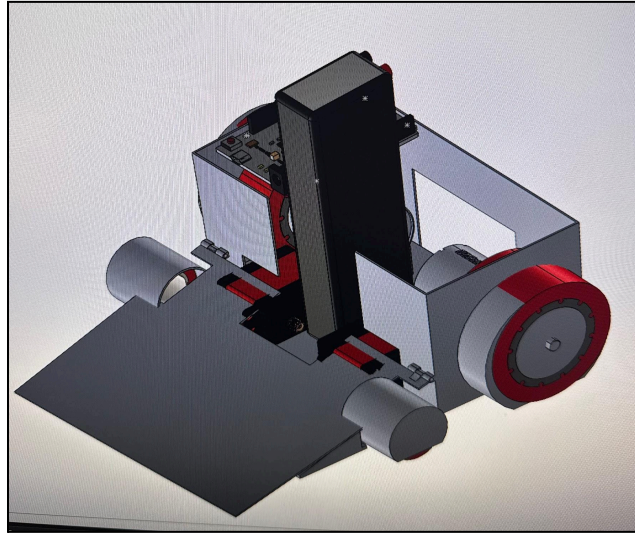


Figure 1: Early CAD model prototype of our components inside a chassis of specified size.
(Created in Solid Works)

Robotics competitions have existed for a long time and thus there is a lot of information online about these robots. Successful sumo robots prioritize traction[5], low center of mass, and prioritize torque over speed or maneuverability. Robots that focus on complex mechanisms over simple advantages can suffer from more issues such as reliability, where more simply designed bots are less prone to failure.

Some previous approaches use four wheel drive systems with high torque motors to maximize their pushing force. High friction tire materials are used to maximize traction against the metal competition ring surface. Minimizing wheel slip is very important. If a robot slips it leads to reduced pushing force or even being pushed out itself.

Chassis geometry also plays a very important role. Low profile designs reduce the likelihood of being lifted or destabilized by an opponent by minimizing the distance a force can act and cause a moment reaction on the robot un-settling it. Many teams use a wedge shaped front end to attempt to gain an advantage. Competition rules restrict ramps, limiting this strategy's maximum potential. Teams must identify an effective chassis geometry which will defend and apply sufficient pushing force on an opponent.

Teams are required to have at least one moving offensive or defensive mechanism in this competition, so teams must navigate the possibilities and pair a strong or complimentary active system with their passive design. A wedge, varying side geometry, asymmetric design, and low surface area would all be examples of passive systems. An active system would use a motor or spring mechanism to move something on the bot to push the opponent, like a flap or arm. Passive mechanisms are easier to integrate than active ones. Active mechanisms introduce complexity and introduce failure modes, but could provide more of a tactical advantage if designed well.

In the context of this project, many existing solutions are not directly transferable due to different vehicle classes or design philosophies. Many parts of existing robots can be re-formed into a new and compliant design[6]. Additionally, the limited development timeline limits the experimentation and re-design process. While some professional robots may rely on advanced materials and expensive electronic components, the competition during the Engineering Expo will be among teams with the same budget and schedule constraints. So we expect the other bots in competition to be of similar parts we have access to.

Our design process is focused around rapid prototyping and revisions. We meet each week to work on aspects of our prototype for the following week. We try to have a new chassis 3D modeled and 3D printed for every Monday meeting so we can place our components inside to make sure they fit and so we can plan revisions. This approach to prototyping helps us quickly make new design decisions and modifications.

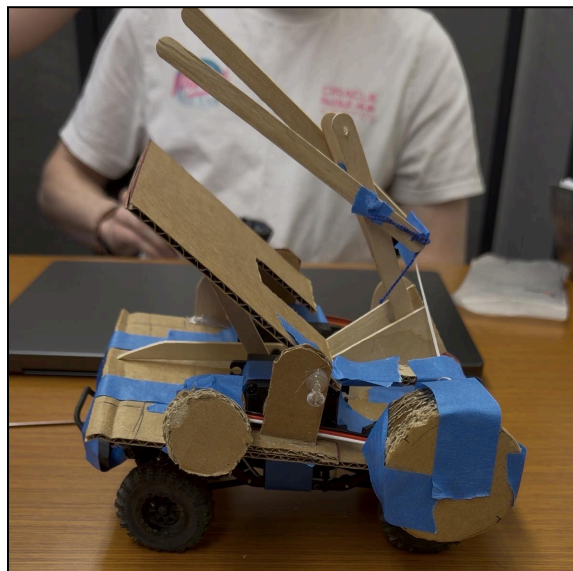


Figure 2: Early prototype showing ramp and claw concept mechanism

The striking feature component of this design will be an intimidating ramp and claw mechanism (see Figure 2, lower ramp made of cardboard and upper claw made of popsicle sticks). On the front of the design will be a ramp which can be actuated up and down. Above this ramp will be two claw like features that can also be actuated up or down. The ramp is intended to slip just underneath an opposing robot and lift it, either slightly to reduce their traction so we can push them off, or fully into the air so we can set them outside the ring. The claw mechanism on top is the key part of this design. The claw will come down to pinch or hold a robot in the ramp mechanism so we can overpower them. When working together, these features of the design will be a force to be reckoned with. One without the other would not be anywhere as strong, but when both are used in harmony the competition stands no chance.

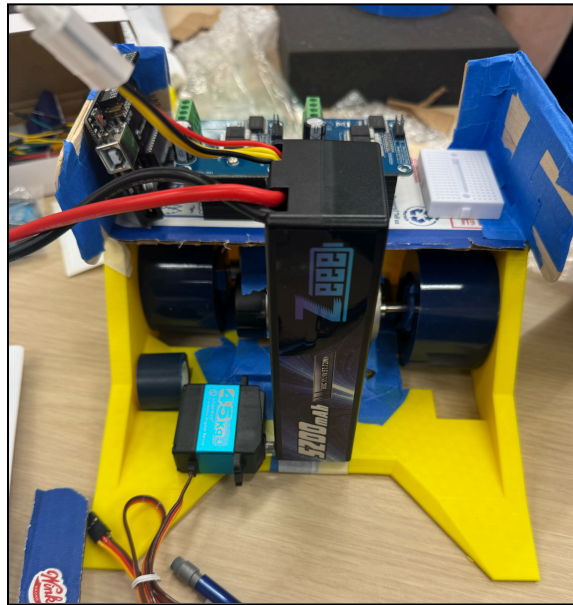


Figure 3: Proof of concept physical prototype chassis with components installed. Assortment of materials used, including the main 3D printed base support, cardboard and popsicle sticks.

The proof of concept shown in Figure 3 was our initial design showing how our components would fit inside the bottom half of our chassis. We learned a lot from this process, mainly we decided where we were going to place our components in the next iteration. The big stressor was the battery which we decided to place upright in this model. After performing another proof of concept we found that we could design a shelf like structure to hold components more efficiently and we decided to lay the battery flat on the third shelf. These shelves will help us keep the components and wires close to each other and minimize our robots surface area. The concept in Figure 3 greatly helped us iterate on our design and 3D model with a new version with the shelves and component spacing.



4 Design Solution

The final design of our SumoBot Blue Mantis represents the collaboration between our team with mechanical, electrical, and software integration to create a unique, rule-compliant design that can perform competitively and matches all customer requirements. The team's approach emphasized system reliability through multiple prototypes and getting creative with grey areas in the robogames rules [1]. This section details the implemented design, its functionality, the compliance with the customer requirements (CRs) and engineering specifications (ESs) along with testing results.

4.1 Description of Solution

The final robot design features a low profile PLA chassis with an aluminum ramp and claw arms that will be utilized together as our offensive feature and pushing strategy. The ramp acts as a way to get under the opposing bot which will then be lifted to destabilize and push using the claws before the opponent can get close to use their ramp. Our robot consists of a four wheel drive train with two high-torque servo motors in the rear and two customized press fitted wheels with an igus rod in the front. Silicone rubber wheels specifically designed for sumobot competition with a friction coefficient of 1.75 were selected for all four wheels to maximize our traction [7].

For the ramp, a dual servo system was integrated to accommodate more torque and a higher distributed weight limit to be sustained on the aluminum ramp with custom servo arms to be attached to the servo. As for the claws, they were manufactured using aluminum with plastidip on the front tips to apply more grip against the opponents. All the servos being utilized are connected to an arduino uno which is all controlled by a Flysky RC receiver transmitter, fulfilling the RC controlled CR 2. With all the electrical components, we found it easiest to create a Printed Circuit Board (PCB) to house all our connections on one board. This also created easier accessibility as the wires are not as bunched up and disorganized as it would be without this feature. To create more accessibility, it was decided to create an acrylic backplate that can slide on and off for an easy access point to the wiring while also adding a cool aesthetic of seeing the inside.

Heavy manufacturing support was provided by our manufacturing sponsor Curtiss-Wright which supplied us with multiple component iterations including two claw and ramp variations, four chassis models, ramp servo attachments, press fitted igus rod customization to front wheels, and multiple claw servo arm attachment spares. These iterations allowed the team to validate dimensional accuracy and mechanical performance for prototypes, final design, and future improvements.

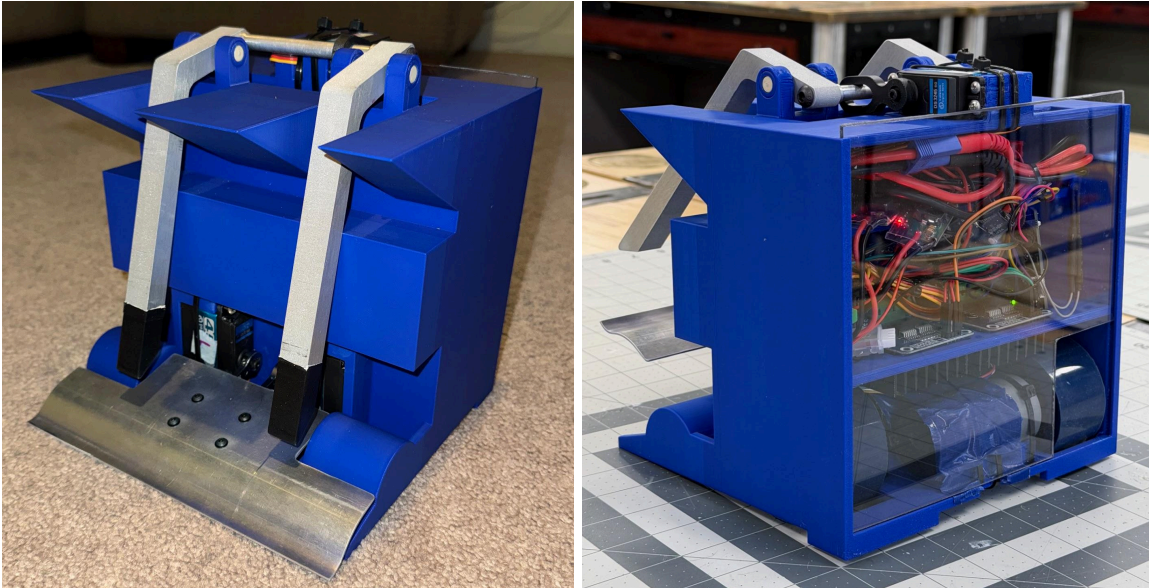


Figure 4: Current Iteration

4.2 Project Results

Through testing, our sumobot demonstrates how all requirements from the house of qualities (Located in Appendix A) are met along with where improvements can be made. Satisfying ES 1, the current sumobot lies within the 20x20 cm footprint at about 20x19cm cutting quite close. Weighing about 2.8kg, there is still room for about 0.2kg to fully optimize the 3kg max weight limit which our team plans to add miscellaneous bearings around the bot to get closer to the max leaving no room for weighing less than opponents. To fully satisfy CR 1, the geometry becomes complex through the claws and supporting triangle features around the claws along with the battery compartment. These areas also satisfy CR 7 with the offensive feature of the claws.

As previously mentioned, the sumobot is controlled via an RC receiver transmitter satisfying CR 2. With 2 high torque motors, 3 high torque servos, and all the weight, we required a big battery which was tested at full power for about 50 minutes. The battery lasted for the full 50 minutes and was only drained about less than half which means it will last through a full competition. The charging speed was also tested in which 30 min charged about half the battery. Our team has two long lasting powerful batteries that can be rotated throughout the expo day satisfying CR 3. To create access to the battery and wiring, a clear backpiece was added that can slide in and out. This feature satisfies CR 4 and also has room for improvement by attaching the same concept to the front side where the battery lies.

Minimal testing for the remaining CRs were conducted however through one pushing force testing observation, it was recorded that our robot applies a 15.89N pushing force and a theoretical traction force of 29.43N. All together, the bot reaches about 54% of its theoretical traction limit. Overall, the final design achieved the intended functionality and competitive performance, validating the engineering specifications and design process. Though all were not mentioned, the bot functioning, driving, and being able to apply such force demonstrates the



traction, durability, and stability as the bot remains upright without tipping and operates fully as intended.

The final price of the project came up to about \$841.99 not including the materials, time, and labor from our Curtiss-Wright sponsor. An estimated cost from Curtiss-Wright came up to about \$3,227 further broke down in Appendix B. This price may be slightly inaccurate with the Chassis price breakdown though \$1,455 of the balance from the rest of the manufacturing remains accurate.

5 LOOKING FORWARD

The next steps for this sumobot project would be to conduct thorough testing to develop new improvements to the claw and ramp functionality. Since these subsystems are dynamic, there are many factors that are hard to account for during the design stage and require more testing to determine strength and functionality. Testing was the area least visited by the SUMO Blue design team, largely due to impending deadlines and limited availability toward the end of the term. Future teams or stakeholders should prioritize designating lots of time. Since this project has moved into the final stages it will be easier to iterate after testing has been conducted. The insights gained from testing proved to be some of the most valuable design feedback the team received.

The current design is functional, but there are small improvements to be made across all dynamic components to ensure a better and consistent output every time. The most noticeable area of concern is the structural support surrounding the claw servo operation. This was addressed to the best of the team's capabilities in the last chassis iteration through the addition of a support beam and underside triangular supports, which were added to give the servo mounting plate greater rigidity and reduce the internal forces acting on it during operation. This iteration came directly out of the team's first rounds of testing, and with more time, further refinement will most likely be needed to confirm that the claw functionality is fully reliable under competitive conditions. The servo moment arm for the claw could also be improved with either custom fine detail CNC machining, or a redesign to reduce the inbetween space in the slot to ensure a direct connection to the servo. We see a bit of play in the current arm when switching directions of the servo rotation.

From a code and controls standpoint, the current system works reliably, but future teams could look into tuning the sensitivity from the received transmitter signals for more precise motor and servo control.

Overall, the SUMO Blue bot is a strong foundation for any iteration testing. With dedicated testing time and continued refinement of all dynamic systems, a future team could easily take this design and produce an even better sumo bot.



Figure 5: Final Chassis

6 CONCLUSIONS-Jonah

The ME-037 sumo bot project of the MIME 497 class challenges Oregon State University ME students to design a sumo robot. The robots in competition will follow the Robo Games Unified Sumo Robot rules[1] in the 3kg weight class. The objective is to develop a robot capable of pushing an opposing robot out of the competition ring.

Research into how previous sumo bot designs chose to solve this problem shows the importance of traction and chassis geometry. By following proven design decisions while adapting and improving aspects of the design, the team hopes to improve competitive performance and build a spectacular robot. Clear identification of the scope, research, stakeholders, constraints, risks, and assumptions for this project will set the team up for success early.

We've identified some important design principles related to the safety and economic factors of our project to follow. We have determined that during the assembly process we are at risk of minor bodily injury, so we will follow PPE standards such as wearing glasses or gloves when cutting, soldering, and other minor risk activities. We also have a budget for this project and have taken measures to ensure we stay below it and purchase the best value products for our design. We have done this by constantly updating the bill of materials we made and doing ample research on products.

Successful completion of this project will reveal a competition ready sumo robot that performs as intended, represents The School of MIME and the Capstone program, provides excellent hands on engineering experience, and prepares members for a career in mechanical engineering.



7 REFERENCES

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8 APPENDICES

8.1 Appendix A: Charts Utilized

CR/ES House of Quality Table

House of Quality

MIME 497

Team Number: MIME 037

Team Description: SumoBlue

Note: 10 rows provided but you may add more as needed or not all of them need to be used.

Customer Requirements (CRs) to Engineering Specifications (ESs)				
CR#	CR description using complete sentences	Weight (100 total)	Matching Engineering Specification	Targets with Tolerances
1	Complex Geometry/ Conforms with maximum size and weight requirements	25	3 kg weight limit 20 cm by 20 cm footprint	Weight: $\leq 3\text{kg}$, Footprint: $\leq 20\text{cm} \times 20\text{cm}$
2	RC Controlled	10	Control input sources	0 sensors used
3	Battery Longevity	15	Last through the full competition duration (3min matches, 4 matches, extra on/off power consideration throughout day)	> 2hr total runtime, 2hr $\pm$ 1 hr for recharge time
4	Easy accessible interior	5	Maintenance access hole for a hand and/or tool with minimum fasteners	Hole diameter >10cm, Uses ≤ 4 fasteners
5	Must be ready to compete in Week 7 of second term	10	Full assembly and functionality completed by deadline	Deadline: 5/15/26
6	Traction	10	High coefficient of friction	Robot doesn't move under 3kg applied push force
7	Offensive/ Defensive Feature	5	Mechanism effectiveness	Force > Weight of opponent bot
8	Durability	10	Structural strength	FEA analysis stress $\leq 0.5 \times$ (yield strength)
9	Stability	10	Low center of mass height	Center of mass < 25% of Height
Sum (should be 100)		100		

FMEA Table

Team Number: 037		Team Name: Sumo Blue	
Part # and Functions	Potential Failure Mode	Potential Effect(s) of Failure	Severity (S)
Front ramp	deflection/ breakage	hinders driving, unable to use ramp, must be removed and leaves front of robot open	4
Claws	deflection/ breakage	unable to grab opponents, must be removed	2
Wheels	slip/ silicon tears	unable to move, easily pushed out of ring	9
Motors	overheating/ low torque	Unable to move, affects other components	10
Battery	runs out of charge/ not supplying enough voltage	robot shuts off, unable to compete	10
RC receiver/transmitter	runs out of charge/ disconnects from controller	unable to control robot, actions are delayed or sporadic	9
Chassis integrity	not substantially strong/ crumbles	robot is completely destroyed and unable to hold itself together	10
Servos	does not supply enough leverage	unable to move ramp or claws	4
Arduino code	bug in the code	robot controls are not interpreted as expected, robot performs undesired or no action	6

Date: 3/11/25					
Potential Causes and Mechanisms of Failure	Occurance (O)	Current Design Controls Test	Detection (D)	RPN	Recommended Action
repeated impacts, weak material use	1	FEA analysis	4	16	bring duct tape
repeated impacts, weak material use	2	FEA analysis	5	20	bring duct tape
falls off the competition ring, sliding sideways	3	pre competition full system shakedown	5	135	consider purchasing spares
over use, poor insulation	2	pre competition full system shakedown	7	140	run at full power and measure temperature change
over use, poor charger	1	pre competition full system shakedown	3	30	run robot for competition length and test battery charge time
interference, faulty wiring, sketchy code	3	pre competition full system shakedown	1	27	ensure wires are secure in testing
repeated impacts, weak material use	1	FEA analysis	2	20	test prototype durability, bring ductape
insufficiently powered	4	pre competition full system shakedown	3	48	ensure robot functions without servo use
inexperienced coding, unsufficient testing	3	pre competition full system shakedown	4	72	bring usb adapter to emergency fix arduino code

Standards of Practice as Applied to this Project

Standard Number or Code	Title of Standard	How it Applies to Project
ANSI/RIA R15.06	Industrial Robot Safety Standard	Provides general safety guidance for robotic systems, including safe operation and control
ASME Y14.5	Dimensioning and Tolerancing	Ensures proper mechanical fit and clear communication of part geometry in CAD drawings
ASTM A36	Standard Specification for Structural Steel	Guides material selection for chassis components requiring strength and durability
ASTM D2000	Rubber Products in Automotive Applications	Applies to wheel material selection for traction and wear resistance
IEEE 802.15	Wireless Communication Standards	Relevant for RC communication reliability and interference considerations
ISO 2768	General Tolerances	Defines acceptable dimensional tolerances for manufactured parts
SAE J1747	Electric Vehicle Battery Safety	Provides guidance for battery handling, charging, and safety
IFI Fastener Standards	Mechanical Fastener Standards	Ensures proper selection and use of bolts, screws, and fastening methods
NIST Engineering Handbook	Measurement and Testing Standards	Guides accurate measurement, calibration, and validation during testing
Unified Sumobot Rules	All Sumo Robot Competition Rules	Defines size, weight, and operational constraints required for competition compliance

Pugh Chart

	CV 1	CV 2	CV 3	CV 4	CV 5	CV 6	CV 7	CV 8	CV 9	CV 10	CV 11	CV 12	CV 13	CV 14	CV 15	CV 16	CV 17	CV 18	CV 19	CV 20
Criteria	Concepts	Ramp	Downward spike	Jumping mechanism	Scissor lift mechanism	Trike lift	3D printed chassis	Machined aluminum chassis	Resin chassis	Carbon fiber chassis	Game RC Controller	Wheel RC Controller	Rubber wheels	Treads	Magnetic wheels	Upward blowing fan blade				
Leverage																				
Traction		-	S	S	S		+	+	+		-		+	+	+					
Low Center of mass		-	-	+	+		+	+	+		S		+	+	+					
Durability		S	-	-	-		+	+	+				S		S					
Stability		-	-	S	-		+	S	+		-		+	+	+					
Complex geometry		+	+	+	+		-	S	+		+		+	+	+					
Offensive		+	-	+	+		S	S	+		-		-	+	+					
Defense		+	+	+	+		+	+	+		+		+	+	+					
Battery life		S	-	-	-		-	S	+		S		-	-	-					
High torque motor		S	-	-	-		S	S	+		S		S	-	-					
Control		-	-	-	+		S	S	-		-		-	+	+					
Cost		-	-	-	-		-	S	-		+		-	S	-					
Assembly/Installation		-	-	-	-		-	S	-		-		-	-	-					
Weight		S	-	-	-		-	S	+		S		-	-	-					
Size		S	-	S	-		S	S	-		-		-	S	S					
$\Sigma(+)$		0	2	2	5	6	0	5	4	10	0	3	0	7	4	8	0	0	0	0
$\Sigma(-)$		0	8	12	12	1	0	8	11	2	8	0	7	0	0	5	0	0	0	0
$\Sigma(S)$		0	5	1	3	1	0	3	11	3	0	4	0	1	0	2	0	0	0	0
DATUM																				
Best Option																				

Decision Matrix

		CV 1		CV 2		CV 3		CV 4		CV 5		CV 6		CV 7		CV 8		CV 9		CV 10					
Description		Ramp		Scissor lift mechanism		Trike lift		3D printed chassis		Resin chassis		Game RC Controller		Rubber wheels		Treads		Magnetic wheels		Upward blowing fan blade					
		Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score				
Criteria	Weight																								
Leverage	10	3	30	4	40	5	50	2	20	2	20	1	10	2	20	2	20	4	40	4	40				
Traction	7	3	21	1	7	2	14	2	14	2	14	1	7	4	28	5	35	5	35	5	35				
Low Center of mass	6	4	24	2	12	4	24	3	18	3	18	1	6	3	18	4	24	4	24	3	18	Excellent	5		
Durability	5	4	20	3	15	4	20	3	15	4	20	1	5	3	15	4	20	4	20	3	15	Average	4		
Stability	4	4	16	2	8	4	16	3	12	4	16	1	4	3	12	4	16	4	16	4	16				
Complex geometry	2	3	6	4	8	4	8	3	6	4	8	3	6	3	6	4	8	4	8	4	8				
Offensive	5	4	20	5	25	5	25	3	15	4	20	4	20	4	20	3	15	3	15	4	20	Not good	1		
Defense	5	4	20	3	15	4	20	3	15	4	20	4	20	3	15	4	20	4	20	4	20				
Battery life	8	5	40	2	16	1	8	5	40	5	40	3	24	4	32	2	16	2	16	1	8				
High torque motor	8	5	40	2	16	1	8	5	40	5	40	4	32	3	24	2	16	2	16	1	8				
Control	5	4	20	2	10	3	15	1	5	1	5	5	25	3	15	3	15	2	10	2	10				
Cost	10	5	50	2	20	2	20	5	50	3	30	2	20	4	40	3	30	2	20	1	10				
Assembly/Installation	5	5	25	2	10	1	5	5	25	4	20	4	20	4	20	3	15	3	15	2	10				
Weight	10	3	30	3	30	3	30	4	40	3	30	1	10	4	40	3	30	2	20	3	30				
Size	10	2	20	2	20	2	20	1	10	1	10	3	30	3	30	1	10	2	20	1	10				
			0		0		0		0		0		0		0		0		0		0				
TOTAL	100		382		252		283		325		311		239		335		290		295		258				
RELATIVE RANK			1		9		7		3		4		10		2		6		5		8				
		Best but average				Second best				Best but average				Only RC option				Best but average				Second Best			



8.2 Appendix B: Budget Sheet

Bill of Materials (OSU Budget)

Bill of Materials										
Item #	Subsystem	Part Name	Description	Supplier	ASIN or part #	Material Specs	Quantity	Unit cost	Extended cost	Notes
1	Motor & Actuation	131.1 Metal Gearmotor 370x73L mm 12V with 64 CPR Encoder (Helical Pinion)	Motors for the back wheels	Pololu	4757	12V, 131.25:1 Gear Ratio, 76 RPM, with Encoder	2	70.95	141.9	
2	Wheels	FingerTech Urethane Sumo Wheel (single)	Back Wheels	FingerTech Robotics	NA	2.50" Dark Blue, Shore A20	2	19.51	39.02	
3	Wheels	FingerTech Mini-Sumo Wheels (pair)	Front Wheels	FingerTech Robotics	NA	Blue, Shore A20	1	19.77	19.77	Pack of 2
4	Wheels	FingerTech Sumo Wheel Hub (single)	Wheel Hubs for the back wheels	FingerTech Robotics	NA	6mm	2	4.64	9.28	
5	Control & Electronics	Flirsky FS-6X 6-10(Default 6)CH 2.4GHz AFHDS RC Transmitter w/ FS-JA66 Receiver	Transmitter Controller	Amazon	B0744DPPL8	Plastic, 6-10 CH 2.4GHz	1	59.99	59.99	Model number FS-6X
6	Wheels	M10 x 16mm Stainless Steel Hex Socket Spring Ball Point Grub Screws 10pcs	Front Ramp Ball Wheels	Amazon	B083DB24YJ	Stainless Steel, M10 X 16 mm	1	6.99	6.99	Pack of 10
7	Power	Zeeo 3S Lipo Battery 5200mAh 11.1V 80C Hard Case RC Lipo with EC5 Connector for 1/5 1/8 1/10 RC Car Boat Truck Helicopter Airplane UAV Drone FPV Racing Models 2 Packs	Battery	Amazon	B0FWB748K8	5200mAh, 3s Lipo, 11.1V	1	70.99	70.99	Pack of 2
8	Control & Electronics	2-Pack BTS7960 43A H-Bridge Motor Driver - High-Power DC Motor Controller with Current Limiting/Overheat Protection for Smart Cars/Robotics/Industrial Automation/Arduino Projects	Motor Driver	Amazon	B0F8F4P7S4	43A H bridge Motor Driver	3	15.38	46.14	Pack of 2, had faulty driver in initial order (only need 2). Burnt out a set.
9	Control & Electronics	ROJACK 8PCS 803 Tie Points and 1 Pcs 400 Tie Points Solderless Breadboard and 65 Pcs Flexible Breadboard Jumper Wires and 6PCS 170 Points Mini Solderless Breadboard	Board with Wires	Amazon	B09Y1LTF1R	Solderless Breadboards, Flexible wires of varying sizes	1	8.99	8.99	Variety Pack of breadboards and wires
10	Tools/Assembly	Zip Ties Assorted Sizes 4"x6" 6"x12" 400 Pack Black Cable Ties, UV Resistant Wire Ties by ANSON	Zip Ties to hold wires together	Amazon	B0C2ZAL3S6	4", 6", 8", 12" plastic zip ties, Tensile strength 45lbs	1	9.99	9.99	Variety Pack with multiple Sizes
11	Motor & Actuation	45KG High Torque Servo 8.4V Coreless RC Steering Servo Motors Stainless Steel Gear Low Profile Waterproof for 1/8 1/10 1/12 Scale Cars CARS DIRT ROBOTIC - 180 Degree	Servos for the ramp and claw arm	Amazon	B0CCPDY8DQ	45 kg/cm, 180 deg, 8.4V	3	31.99	95.97	
12	Power	FEICHAO 8A UBEC Output 5V / 6V 8A Max 12A Input 7V-25.5V 2-6S Lipo / 6-16 Cell Ni-MH Input Switch Mode BEC for RC Helicopters	UBEC for servo	Amazon	B07DD9L6P6	Input: 7V-25.5V (2s-6s Lipo), output voltage: 5V/6V, Current 8A max 12A	4	18.98	75.92	Only need 2 but 2 had fried needed replacement
13	Power	2PCS RC BEC UBEC 5V 5A Step Down UBEC Module 2S 3S 4S for FPV Quadcopter Airplane RC Servo RC Car Boat Motor	UBEC for arduino	Amazon	B0D97FHJHS	Input: 5.5V-35V (2-7-cell Lipo), output voltage: 5V/5A, Current 8A	2	9.99	19.98	Pack of 2, first pack fried
14	Power	Battery Charger for 2S 3S Lipo Battery, Balance Charger for XHR Plug Battery, Lipo Chargers for RC Toy & Model, RC Battery Chargers with Smart Chip	Battery Charger	Amazon	B0FPDJ5057	Input: 120V, Output 7.4V, for 2s 3s Lipo Battery, XHR Plug	1	16.99	16.99	
15	Control & Electronics	Sink-Lon Dupont Wire Breadboard Jumper Wires Jumper Wire Male to Female for Arduino Breadboard and Circuit Board (5m 4 Colors Total 40-Pack/Male to Female)	Male to Female Wires	Amazon	B08M3QLL3Q	6cm long wires, pack of 40	1	8.99	8.99	Pack of 40
16	Power	HCDC HD062.8 Output DC Power Distribution Block Module for 15/30/45A Connects	Power Distribution Block	Amazon	B0C6KGRQPD	8 Output DC Power, 15/30/45A connects, 60V	1	45	45	
17	Control & Electronics	Ferrymy 14 AWG Stranded Wire Spool 10ft Each 6 Colors Flexible 14 Gauge Silicone Hook up Wire Kit Electrical Tinned Copper Wire	Ferrymy Wire	Amazon	B089CRKSNM	Silicone, Tinned Copper, 14 AWG, 6 spools 10ft each	1	23.09	23.09	Pack of 6 spools
18	Tools/Assembly	Taisi Dupont Crimping Tool Kit with 600PCS 2.54mm Dupont Connector Kit SH-28B Ratcheting Wire Crimper Plier 1/2/3/4/5/6/7 Pin Housing Dupont Connector Male Female Pin Header Crimp Terminal AWG26-18	Crimping Tool Kit	Amazon	B0B11RLGDZ	600 Pcs 2.54mm Dupont Connector kit	1	21.99	21.99	Tool Kit
19	Power	DGZZI 2PCS EC5 Male Connector to 4.0mm Banana Male Plug Lipo Battery Balance Charging Cable 12 AWG Silicone Wire Charger Cable Adapter for RC Helicopter Vehicle Toys 100mm	EC5 Male Connector	Amazon	B07R6RP36W	4.0 mm Banana Male Plug 12AWG Wire	1	6.99	6.99	Pack of 2
20	Tools/Assembly	Duct Tape	Duct tape for fixes between matches on EXPO day	Amazon	B062TOL400	1.88 In x 20 YD	1	3.5	3.5	
21	3D Printed / Machined	Chassis	Body of our robot	Exterio	NA	PLA, Aluminum, or Carbon	1	0	0	Exterior Sponsor Supplying
22	3D Printed / Machined	Ramp	Ramp Mechanism	Exterio	NA	PLA, Aluminum, or Carbon	1	0	0	Exterior Sponsor Supplying
23	3D Printed / Machined	Claw	Offensive Mechanism	Exterio	NA	PLA, Aluminum, or Carbon	2	0	0	Exterior Sponsor Supplying
24	Tools/Assembly	Diamu 68 PCS 6832Z/ 6232Z/ 6242Z/ 6832Z/ 6872Z/ 6272Z Ball Bearings Assortment Kit, Small Bearings Carbon Steel Deep Groove Bearings, for Furniture Wheel, Roller Skates, 3D Printing and ect — 6 Size	Bearings for front wheel axle movement and possibly claw movement	Amazon	B0FPX7KR8V	Stainless Steel, Multi-pack	1	12.99	12.99	Variety pack with multiple sizes
25	3D Printed / Machined	Servo attachments	Attachments for ramp and claw movement from servos	Exterio	NA	Aluminum	3	0	0	Exterior Sponsor Supplying
26	Tools/Assembly	Scotch Vinyl 700 Electrical Tape, 1 Roll, Black, 3/4 in x 66 ft, General Purpose, Commercial Grade, Rated for Temperatures Up to 194-Degree, Highly Flexible, Stretchy & Conformable (700)	Electrical tape for wire safety and fixtures	Amazon	B001AXDOEY	3/4 in x 66ft	1	2.98	2.98	
27	Tools/Assembly	3D Printer Super Glue Thin for PLA, ABS, PETG, TPU & Nylon - 1.76 oz XXL - Capillary Action, Ultra Fast Drying & Clear - Perfect for Resin, RC Parts, Cracks & Fine Detail Fixes	3D Printer Super Glue	Amazon	B0C8F5MLCF	1.76oz	1	11.99	11.99	
28	Control & Electronics	Chanzon 40pcs 30cm Long Male to Female Header Jumper Wire Dupont Cable Line Connector 40 pin Solderless Multicolored for Arduino Raspberry pi Electronic Breadboard Prototype PCB Board	Male to Female Wires, Extra length	Amazon	B09FPM9J5S	30 cm in length, pack of 40	1	8.49	8.49	Pack of 40
29	Tools/Assembly	3mm x 300mm 304 Stainless Steel Rod 12PCS 3mm Rod, Metal Rods for Crafts Round Solid Shaft Rods Lathe Bar Stock for DIY Car	Stainless Steel shaft to be used as axle for both front wheels	Amazon	B0F8R61RNV	Stainless Steel Shaft, 3mm diameter, 300mm length	1	7.99	7.99	Pack of 12
30	Tools/Assembly	58Pcs 6 Sizes Double Sided PCB Circuit Prototype Solder Board Kit with 2.54mm 40pin Male Female Header Connector and 5mm XF301 Screw Terminal Block for DIY Soldering	DIY PCB Solder board kit	Amazon	B0CKZ7GFSK	58 Pcs, 6 sizes, 2.54mm 40 pin male to female connector	1	13.99	13.99	Tool Kit
31	Tools/Assembly	Amazon.com 5 Pack ShareGoo 25T Aluminum Servo Horns M3 Threads Metal Steering Arm for RC Car Truck Ruggy Airplane-Black - Toys & Games	Servo Arms	Amazon	B07DRNSYLG		1	9.92	9.92	
32	Tools/Assembly	Screwdriver Set	Screwdriver Set	Ace Hardware			1	5.83	5.83	
33	Tools/Assembly	Screws	Screws	Ace Hardware			1	8.59	8.59	
34	Tools/Assembly	Lexan Polycarbonate Sheet (Back piece)	Lexan Polycarbonate Sheet (Back piece)	Home Depot			1	11.98	11.98	
35	Tools/Assembly	Plasti Dip, For claw grip	Plasti Dip, For claw grip	Home Depot			1	7.98	7.98	
36	Tools/Assembly	Plastic Cutter	Plastic Cutter	Home Depot			1	7.78	7.78	
Total Cost:									841.99	

Curtiss-Wright Cost
(Chassis was being double checked, may be incorrect)

Component	Price	Quantity	Total
3D Printed Chassis	\$ 443.00	4	\$ 1,772.00
3D Printer Servo Arms	\$ 17.50	20	\$ 350.00
Claw, Alum	\$ 135.00	2	\$ 270.00
Claw, 3D Printed	\$ 125.00	2	\$ 250.00
Ramp, Alum	\$ 110.00	2	\$ 220.00
Pivot Axle, Claw	\$ 85.00	1	\$ 85.00
Axle, Wheel	\$ 65.00	2	\$ 130.00
Lift Bar, Alum	\$ 75.00	2	\$ 150.00
Grand Total			\$ 3,227.00